EL3 Worked Examples: Introduction to Logarithms

Recording companion for the EL3 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What power do I raise the base to in order to get the number inside the logarithm? That power is the value of the logarithm. An exponential equation and a logarithmic equation say the same thing in two different ways.

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Example 1: Convert 1

$$3^4=81$$

Question

Which equation is the logarithmic form of this exponential equation?

- **A.** $\log_3 4=81$
- **B.** $\log_4 81=3$
- **C.** $\log_3 81=4$
- **D.** $\log_{81} 3=4$

Answer and reasoning

The base of the exponential, 3, becomes the base of the log. The exponent, 4, becomes the value of the log, and 81 goes inside. Read it back as a check: $\log_3 81=4$ says "3 to the 4th power is 81." **The logarithmic form is $\log_3 81=4$.**

Example 2: Convert 2

$$7^n = p$$

Question

Which equation is the logarithmic form of this exponential equation?

- **A.** $\log_7 n = p$
- **B.** $\log_p 7 = n$
- **C.** $\log_7 p = n$
- **D.** $\log_n 7 = p$

Answer and reasoning

Letters follow the same pattern as numbers. The base, 7, stays the base. The exponent, n, is the value of the log. The result, p, goes inside the log. Read it back: $\log_7 p = n$ says "7 to the n is p." **The logarithmic form is $\log_7 p = n$.**

Example 3: Convert 3

$$\log_5 125=3$$

Question

Which equation is the exponential form of this logarithmic equation?

- **A.** $3^5=125$
- **B.** $5^{125}=3$
- **C.** $5^3=125$
- **D.** $125^3=5$

Answer and reasoning

The base of the log, 5, is the base of the power. The value of the log, 3, is the exponent. The number inside, 125, is the result. Check it: $5\cdot 5\cdot 5=125$. **The exponential form is $5^3=125$.**

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Example 4: Convert 4

$$\log_b a=c$$

Question

Which equation is the exponential form of this logarithmic equation?

- **A.** $a^c=b$
- **B.** $b^c=a$
- **C.** $c^b=a$
- **D.** $b^a=c$

Answer and reasoning

Start at the base, b. The value of the log, c, is the exponent. The number inside, a, is the result. Read it back: $b^c=a$ says "b to the c is a." **The exponential form is $b^c=a$.**

Example 5: Evaluate 1

$$\log_2 32$$

Question

What is the value of this logarithm?

A. 16
B. 64
C. 5
D. 2

Answer and reasoning

Ask: 2 to what power gives 32? Count the factors: $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$, so $2^5 = 32$. The value of the logarithm is that exponent. **$\log_2 32 = 5$.**

Example 6: Evaluate 2

$$\log_3(1/9)$$

Question

What is the value of this logarithm?

A. 2
B. −2
C. −9
D. *1/2*

Answer and reasoning

Ask: 3 to what power gives *1/9*? Since $3^2=9$, a negative exponent flips it to $3^{-2}=1/9$. A logarithm can be negative when the number inside is between 0 and 1. **$\log_3(1/9)=-2$.**

Example 7: Evaluate 3

$$\log_{16} 4$$

Question

What is the value of this logarithm?

- **A.** 4
- **B.** 2
- **C.** *1/2*
- **D.** *1/4*

Answer and reasoning

Ask: 16 to what power gives 4? A square root undoes a square, so $16^{1/2}=\sqrt{16}=4$, and the power is *1/2*. Do not flip the base and the number: $\log_4 16$ would be 2, but here the base is 16. **$\log_{16} 4=1/2$.**

Example 8: Common Log

$$\log 0.01$$

Question

What is the value of this logarithm?

A. 2
B. −2
C. −100
D. 0.1

Answer and reasoning

With no base written, the base is 10. Ask: 10 to what power gives 0.01? Since $0.01=1/100=1/(10^2)=10^{-2}$, the power is -2. **$\log 0.01=-2$.**

Example 9: Natural Log 1

$$\ln 7 = x$$

Question

Which equation is the exponential form of this logarithmic equation?

- **A.** $e^{7}=x$
- **B.** $7^{x}=e$
- **C.** $e^{x}=7$
- **D.** $x^{e}=7$

Answer and reasoning

ln is a logarithm with base e, so $\ln 7=x$ means $\log_e 7=x$. The base is e, the exponent is x, and the result is 7. **The exponential form is $e^{x}=7$.**

Example 10: Natural Log 2

$$\ln(e^5)$$

Question

What is the value of this logarithm?

A. e
B. e^5
C. 5
D. $1/5$

Answer and reasoning

Ask: e to what power gives e^5? The power is 5. The natural log undoes a power of e.
$\ln(e^5)=5$.